

Task 3 – Weather App (API Integration Project)

Project Title

Weather App using Public Weather API

Project Description

The Weather App is a web application developed using HTML, CSS, and JavaScript that fetches real-time weather information from a public weather API. Users can search for a city to view the current weather conditions, including temperature, humidity, wind speed, and weather description. The application uses the Fetch API to retrieve live weather data and displays loading and error messages to provide a better user experience.

Objectives

- Integrate a public REST API into a web application.
- Fetch and display real-time weather information.
- Handle loading and error states effectively.
- Create a responsive and user-friendly interface.
- Implement a city search feature for dynamic weather updates.

Technologies Used

- HTML5
- CSS3
- JavaScript (ES6)
- Fetch API
- Public Weather API (OpenWeatherMap)

Features

- Search weather by city name.
- Displays current temperature.

- Shows humidity and wind speed.
- Displays weather conditions (e.g., Clear, Cloudy, Rainy).
- Loading indicator while fetching data.
- Error message for invalid city names or network issues.
- Responsive design for desktop and mobile devices.
- Dynamic weather updates using API integration.

API Used

- **OpenWeatherMap API**

Purpose:

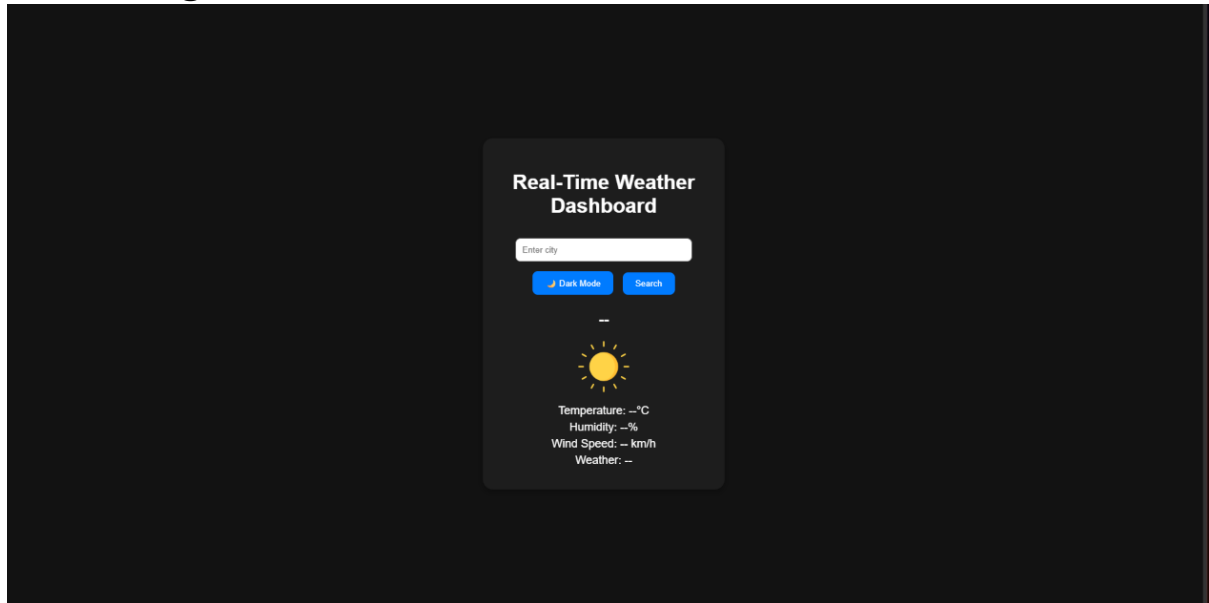
The API provides real-time weather data based on the city entered by the user. The application sends a request to the API using the Fetch API and receives weather information in JSON format, which is then displayed on the webpage.

Project Files

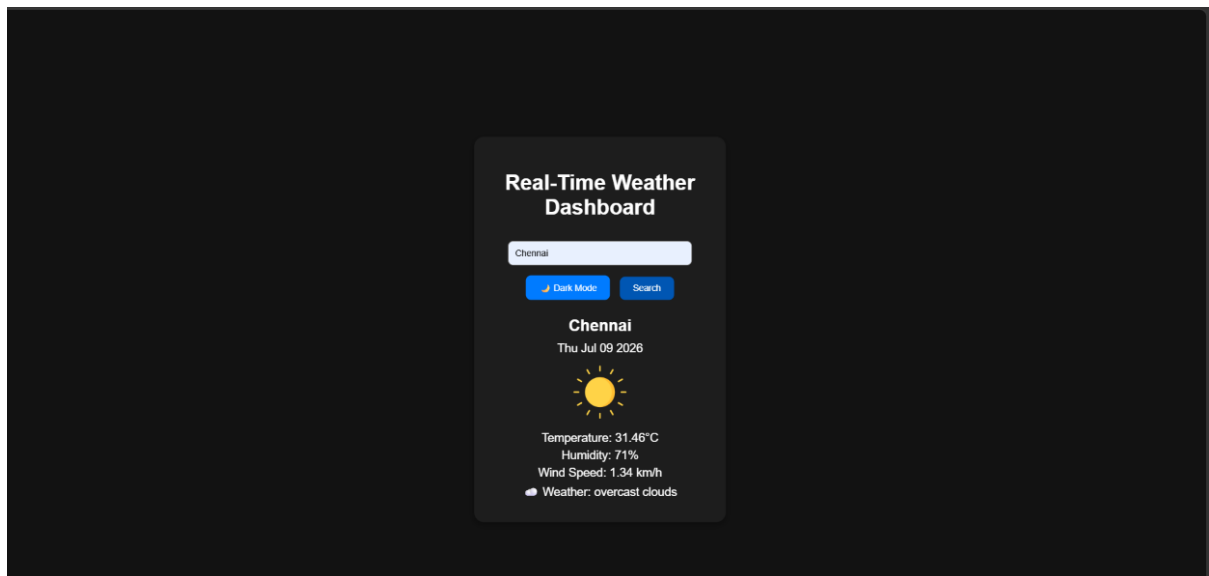
- index.html
- style.css
- script.js

Screenshots

1.Home Page



Result Page



GitHub Repository Link

<https://github.com/Jasminejasy23/task-3-API-INTERGRATION-PROJECT-WEATHER-APP->

Live Demo Link

<https://jasminejasy23.github.io/task-3-API-INTERGRATION-PROJECT-WEATHER-APP-/>

Project Outcome

The Weather App successfully demonstrates the integration of a public REST API into a web application. It fetches and displays live weather information dynamically, handles loading and error states effectively, and provides users with a responsive and interactive experience. This project showcases practical knowledge of API integration, asynchronous JavaScript, and front-end development.

Conclusion

This project enhanced my understanding of API integration using the Fetch API, asynchronous programming, JSON data handling, and DOM manipulation. It also improved my skills in creating responsive and interactive web applications that communicate with external services.